CELLULITIS

Definition: Cellulitis is an acute, spreading, diffuse inflammation of the skin and subcutaneous tissue, usually caused by bacterial infection, without a localized collection of pus in uncomplicated early disease.

1. Etiology and causative organisms

Organism	Clinical relevance
Streptococcus pyogenes (Group A streptococcus)	Classically associated with rapidly spreading, non-purulent cellulitis.
Staphylococcus aureus	Important in wound-associated or purulent infection.
MRSA	Consider when risk factors or purulent infection are present; follow local guidance.
Pasteurella multocida	Animal or cat bites.
Eikenella corrodens	Human bites.
Vibrio vulnificus	Seawater/seafood exposure, especially in susceptible patients.
Mixed aerobic and anaerobic organisms	May occur in diabetic foot, deep wounds and perineal infections.

2. Predisposing factors

• **Break in skin:** cuts, abrasions, wounds, ulcers, burns or insect bites.

• **Skin disease:** eczema, fungal infection between toes and chronic dermatitis.

• **Diabetes mellitus:** impaired immunity and delayed wound healing.

• **Lymphedema or venous insufficiency:** impaired drainage, edema and skin breakdown.

• **Immunosuppression:** HIV, corticosteroid use or chemotherapy.

• **Obesity** and **previous cellulitis** can increase risk of recurrence.

Exam point: Look for a portal of entry, such as an ulcer, wound or fissure between the toes.

3. Pathogenesis

1. Entry of bacteria Cut, wound, ulcer, bite or skin fissure
2. Multiplication In the dermis and subcutaneous tissue
3. Acute inflammation Vasodilation, increased vascular permeability and leukocyte migration
4. Clinical features Redness, warmth, edema, pain and tenderness
5. Possible progression Lymphangitis, lymphadenitis, abscess, bacteremia or sepsis

Inflammatory mediators cause vasodilation and increased capillary permeability. Fluid and inflammatory cells enter tissues, producing edema, redness, warmth and tenderness.

4. Classification

• **Non-purulent cellulitis:** diffuse inflammation without a drainable collection; often associated with streptococci.

• **Purulent cellulitis:** associated with pus, a boil or abscess; *S. aureus* is an important possibility.

• **By site:** facial, periorbital, limb, hand, diabetic foot and perineal cellulitis.

• **Erysipelas:** more superficial infection of the upper dermis and superficial lymphatics, usually with a well-defined raised margin.

• **Necrotizing soft-tissue infection:** deep, rapidly progressive infection with tissue necrosis; a surgical emergency.

5. Clinical features

Symptoms

• Pain and tenderness over the affected area.

• Redness (erythema), swelling and local warmth.

• Fever with chills, malaise or fatigue in moderate/severe infection.

• Difficulty moving the affected part when pain and swelling are significant.

Signs

• Erythema with poorly defined margins.

• Local warmth, tenderness and diffuse edema (sometimes pitting).

• Skin tension due to swelling.

• Portal of entry such as wound, ulcer, fissure or bite mark.

• Regional lymphadenopathy; red proximal streaks may indicate lymphangitis.

• Fever, tachycardia or hypotension in systemic illness.

Usually absent in uncomplicated cellulitis: a clearly localized, fluctuant collection of pus. Fluctuation should prompt assessment for an abscess, although deep collections may not be fluctuant.

6. Clinical progression

Stage/description	Typical features
Early	Redness, warmth and tenderness.
Progressive	Increasing edema, pain and spreading erythema.
Severe	Fever, tachycardia, marked swelling or systemic toxicity.
Complicated	Abscess, tissue necrosis, bacteremia or sepsis.

These are useful clinical descriptions, not rigid universally accepted stages; not every case progresses through each stage.

7. Investigations

Investigation	Importance
Clinical examination	Primary basis of diagnosis; assess spread, entry wound and severity.
CBC	May show leukocytosis and neutrophilia.
CRP	Assesses inflammation; may help monitor response.
Blood glucose	Detects hyperglycemia and helps assess diabetic patients.
Renal function/electrolytes	Useful in severe infection and before certain treatments.
Blood cultures	Consider in severe systemic illness, immunocompromise or unusual exposures; not routine in every mild case.
Pus culture	If purulent drainage or an abscess is present.
Ultrasound	May identify a hidden abscess when examination is uncertain.
CT/MRI	Selected deep or complex infections; do not delay surgery when necrotizing infection is strongly suspected.

8. Differential diagnosis

Condition	Distinguishing feature
Cellulitis	Diffuse redness, warmth, tenderness and edema; poorly defined margins.
Abscess	Localized collection of pus; may be fluctuant.
Erysipelas	Superficial infection with a raised, sharply demarcated margin.
Necrotizing fasciitis	Disproportionate pain, rapid progression, systemic toxicity, bullae or necrosis.
Deep vein thrombosis	Limb swelling and pain; can mimic cellulitis.
Contact dermatitis	Itching, exposure history and sometimes geometric distribution.
Gout	Acute joint-centred pain, tenderness and swelling.
Septic arthritis	Severe joint pain with restricted passive and active movement.

Red flags: pain out of proportion, rapid deterioration, bullae, skin discoloration, anesthesia or crepitus should raise concern for necrotizing infection.

9. Treatment

Principles

• Assess severity and identify emergencies.

• Start appropriate antibiotics.

• Elevate the affected part.

• Treat the portal of entry and underlying risk factors.

• Drain pus if an abscess is present.

• Monitor clinical response and complications.

Initial assessment

• Check temperature, pulse, blood pressure, respiratory rate and general condition.

• Assess extent and rate of spread, pain severity and tenderness.

• Look for pus, fluctuance, bullae, discoloration, crepitus or deep infection.

• Check distal pulses, sensation and movement when relevant.

• Ask about diabetes, immunosuppression, allergies, recent antibiotics and prior MRSA.

Supportive care

• **Elevation:** elevate the affected limb above heart level when feasible to reduce edema and discomfort.

• **Rest:** rest during the acute painful phase; gradually resume movement as symptoms improve.

• **Analgesics/antipyretics:** paracetamol may be used when appropriate; other medicines depend on contraindications.

• **Hydration:** maintain oral fluids; IV fluids may be required for dehydration, hypotension or systemic illness.

• **Local care:** gently clean and cover wounds; treat ulcers, fungal infection or other portals of entry. Do not squeeze or puncture the area.

Antibiotic therapy

Antibiotics are the main treatment for bacterial cellulitis. Choice depends on severity, likely organisms, allergies, local resistance patterns and comorbidities. These are educational examples, not an individualized prescription.

Clinical situation	General approach
Mild, uncomplicated non-purulent	Oral antibiotic targeting streptococci, often with activity against MSSA.
Moderate infection or unable to tolerate oral therapy	Hospital assessment; IV treatment may be required.
Severe infection/systemic toxicity	Urgent admission, IV antibiotics and assessment for sepsis.

Clinical situation	General approach
Purulent infection or MRSA risk	Drain abscess if present and use appropriate MRSA coverage when indicated.
Bite-associated infection	Cover expected bite-associated organisms.
Diabetic foot/perineal infection	Assess depth and severity; broader coverage may be required.

Examples used in selected settings include flucloxacillin or cephalexin for suitable uncomplicated infections, and IV cefazolin or an antistaphylococcal penicillin for selected hospitalized patients. MRSA-active drugs such as vancomycin are reserved for appropriate indications.

• Antibiotics alone are not enough when a drainable abscess is present.

• Broad-spectrum combinations are not needed for every uncomplicated case.

• Reassess if worsening or not beginning to improve within about 48–72 hours.

• An uncomplicated case that is improving often needs around 5 days of therapy; extend according to response and severity.

Surgical management

• **Incision and drainage (I&D;):** for a confirmed abscess.

• **Debridement:** removal of dead tissue when necrosis or another surgical indication is present.

• **Urgent exploration/debridement:** for suspected necrotizing soft-tissue infection.

• **Source control:** wound management, treatment of an infected ulcer or removal of an ongoing source when indicated.

Why not incise ordinary cellulitis? Uncomplicated cellulitis causes diffuse inflammation rather than a localized collection of pus. Incision without a collection does not provide effective drainage and may injure tissue.

Management according to severity

Severity	Approach
Mild	Clinically stable, no systemic toxicity or deep infection signs: oral antibiotics when indicated, elevation, analgesia and reassessment.
Moderate/high-risk	Consider admission or supervised care for systemic illness, significant comorbidities or inability to take oral therapy; investigate and use IV antibiotics when indicated.
Severe/emergency	Immediate surgical consultation if necrotizing infection is suspected; resuscitation, urgent broad-spectrum IV antibiotics and prompt operation when indicated.

10. Complications

• **Abscess formation:** localized collection of pus.

• **Lymphangitis:** inflammation of lymphatic vessels, often causing red streaks.

• **Lymphadenitis:** inflammation of regional lymph nodes.

• **Bacteremia and sepsis:** bacteria enter blood and may cause life-threatening systemic illness.

• **Necrotizing fasciitis:** rare but serious deep soft-tissue infection.

• **Osteomyelitis or septic arthritis:** possible spread from an adjacent wound or infection.

• **Chronic lymphedema and recurrence:** may follow lymphatic damage or persistent risk factors.

11. Cellulitis vs abscess

Feature	Cellulitis	Abscess
Definition	Diffuse infection of skin/subcutaneous tissue.	Localized collection of pus.
Extent	Diffuse, spreading.	Localized.
Tenderness	Diffuse.	Usually maximal over collection.
Fluctuation	Usually absent.	Often present when superficial and mature.
Treatment	Antibiotics and supportive care.	I&D; when indicated; antibiotics according to clinical circumstances.

12. Cellulitis vs erysipelas

Feature	Cellulitis	Erysipelas
Tissue involved	Deeper dermis and subcutaneous tissue.	Superficial dermis and lymphatics.
Margin	Poorly defined.	Sharply demarcated, raised.
Appearance	Diffuse redness and swelling.	Bright red, raised plaque.
Common organisms	Streptococci and *S. aureus*.	Mainly beta-hemolytic streptococci.

13. Prevention and prognosis

• Keep wounds clean and covered; treat skin breaks early.

• Treat fungal infection between toes, ulcers and chronic skin disease.

• Maintain good diabetic control and appropriate foot care.

• Manage chronic edema and venous insufficiency.

• For recurrent cellulitis, identify persistent risk factors; selected patients may need specialist advice about preventive antibiotics.

• Most uncomplicated cases improve with appropriate treatment. Pain, fever and spread should begin to settle within the first few days; swelling and discoloration may take longer.

Seek urgent reassessment for rapidly spreading redness, worsening pain, persistent fever, blisters, dark skin changes, confusion or faintness.

14. References

1. Williams NS, O'Connell PR, McCaskie AW. *Bailey & Love's Short Practice of Surgery*. 27th ed.

2. Townsend CM, Beauchamp RD, Evers BM, Mattox KL. *Sabiston Textbook of Surgery*. 21st ed.

3. Bhat Sriram M. *SRB's Manual of Surgery*. 6th ed.

4. Brunicardi FC, Andersen DK, Billiar TR, Dunn DL. *Schwartz's Principles of Surgery*. 11th ed.

5. Zinner MJ, Ashley SW, McFadden DW. *Maingot's Abdominal Operations*. 13th ed.